Sprint Report 6

Team Plugged In
David Jarman, Marshall Gaucher, Kevin Worner
Project STBMF
March 20th –April 10th 2013

Initial Sprint Backlog

This sprint focused on finishing up testing and writing documentation. Through unit and load testing, bottlenecks and errors were found and corrected in the server. The STB client has a couple of features to finish up as well.

David and Kevin focused on the server portion while Marshall continued to work on the Virtual Set-Top Box. All three members were involved in documentation and creating the poster.

Server

1. Load testing tool
    1.1. Communication Layer
    1.2. Load static game from file
    1.3. Write results to .CSV file
2. Add better error checking and thread safety to DatabaseConnection
3. Add logic to logger to write to service event log
4. Add mirroring to each player sees their side of the board on the bottom
5. Unit tests
    5.1. Callbacks
    5.2. Callback Extensions
    5.3. Logger
    5.4. CreateGameStateRequest

Virtual STB

1. Finalize virtual remote
    1.1. Use GIMP to chop vstb remote button images
    1.2. Set widget background to main remote image
    1.3. Set pushbuttons to according chopped remote button property
    1.4. Move Up
    1.5. Move Down
    1.6. Move Right
    1.7. Move Left
    1.8. Ensure button stays within borders for a given move button
    1.9. Ok (get current button with set down property set as nSelectedTileIndex)
2. Crashes when exiting from an old game and starting a new one
3. Unit Testing
    3.1. Unit test for model classes
    3.2. GUI test for view
    3.3. Modify Doxygen make file to generate call graph for dependency hierarchy
    3.4. Check if qtestlib will allow test file out to database through provided qt unit test signal/slot mechanism. Ideally it should work its Qt
4. Redo database binds for new log tables
    4.1. Connect to local database

   4.2. Change qDebug statements to qsqlquery statements
   4.3. Ensure tables reflect live insert statements
5. Ensure project follows coding standards

Common

1. Design Document
   1.1. Overview
   1.2. Project Overview
   1.3. Requirements
   1.4. Design and Implementation
   1.5. System and Unit Testing
   1.6. Development Environment
   1.7. Release | Setup | Deployment
   1.8. End User Documentation
   1.9. Appendices
2. Create working live demo for Design Fair
3. Client Presentation 3
   3.1. Create slides
   3.2. Make PDF version
   3.3. Practice presentation as group
4. Ensure STBMF demo for senior design works with LAN setup
5. Design Fair poster

Extra Work Completed

1. Microsoft Fakes – Server unit tests were re-written using 'shims' and 'fakes', an easier way to provide mock objects
2. David converted the server solution to be able to run as a windows service
3. David and Kevin worked on a Load Testing tool to quantify how well the server performed under real stress of many users. This provided valuable data and also helped identify bottlenecks within the software, which were then analyzed and corrected.

Comments and Reflection on Sprint

This sprint was mostly used to test our system, create a poster and prepare for the Design fair. We added more unit tests to the server and have more than 50% code coverage. We changed the way we were mocking objects by moving to a new framework called Microsoft Fakes. This has allowed us to write our tests quicker while making them easier to understand.

Work Left Over

Server

The server does not have a User Interface yet, but this feature was given low priority by the client. It will require more research, but it should not take long to create.

The code still uses a static connection string to talk to the database. This will be refactored later to add a new one at runtime.

Matchmaking will be re-added after the design fair is complete.

Checkers

The A.I. in checkers is still woefully inefficient and not scalable.

Virtual STB

The vSTB still needs to request resources from the server rather than using a static location on the C:\ drive. There is also a Tester card still in the backlog that is unfinished.

Unit testing will also occur in the next couple of weeks that will ensure that the STB will not crash from invalid states that could possibly occur.

Project

Work with Brian to work on a way to deploy the system to other servers, including creating the database and filling it with initial data, loading plugins, and managing the database connection string.

Research/Code Experiments

Microsoft Fakes

While looking for better unit testing libraries, David stumbled upon a relatively new technology from Microsoft. It required some tests to be re-written, but also provided valuable techniques for testing, such as swapping out runtime assemblies for static functions, so that custom functionality can be injected, even in .NET methods and classes.

Artificial Intelligence Research

UCI: Universal chess interface.

Found an AI written in C++. This will give us more ideas and inspiration on creating a better AI.

Meetings

Each week, the team met with Scrum Master Brian Butterfield, and clarified our vision and gave updates as to our project. He would also help clarify on questions we had come up with, and occasionally offer tips as to how to implement certain portions.